

Enhancing and Securing Agricultural Products Traceability Using Ethereum Based AgroTrustChain Framework

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Abstract — The area of agriculture supply chain traceability has become increasingly significant due both to concerns about food safety (including regulation requirements) and increasing demand among consumers to have a clearer understanding of how the food they buy was produced. Traditional blockchains-based traceability systems such as Weighted Two-Hop Raft consensus mechanisms address both consensus scalability issues and supplier trustworthiness issues in large scale agricultural supply chains (also referred to as ecosystems). These supply chain systems partition the suppliers by creating hierarchical groups, and reassign voting weights according to both reputation and product quality. As a result, communication overhead associated with consensus for multi-layers supply chains has been reduced; in addition, the efficiency of obtaining consensus among parties has also been increased through the use of Voting weights assigned dynamically to each supplier during the voting process. By virtue of being decentralized, blockchains can provide immutability, distributed validation and transparent audit trails; therefore they hold the potential for many applications within agriculture which require traceability. The majority of existing implementations utilize permissioned consensus protocols that may enhance communication but do not necessarily provide centerless consensus (i.e. they are not designed to fuel true centerless dcns). In these systems, leader-follower structures and weighting schemes are used in order to minimize "malicious behaviour" from suppliers while still achieving throughput performance. While there are significant benefits to using Weighted Two-Hop Raft compared to classic Raft (such as delay in obtaining consensus and throughput times), the underlying architecture will continue to rely on semi-trusted nodes and will not be capable of automating Smart Contracts without a true decentralised configuration. Overall, there are several major disadvantages associated with the base system. The first principal limitation being that it operates solely in a permissioned blockchain environment; all nodes must be predetermined by the users prior to use and therefore limits the availability and minimisation of trust. The use of leader-based coordination in the second version of the Weighted Two-Hop Raft introduces partial centralization risks and potential bottlenecks when there is a high volume of transactions. It also calculates voting weights based on reputation and quality which means it relies on an external scoring system that adds to the complexity of the calculation and introduces the potential for bias. In addition, the framework does not have programmable automation for enforcing agricultural compliance policies such as certification validation, expiry tracking or automated penalties. Furthermore, there are limitations with respect to interoperability with public blockchain ecosystems which results in limited transparency for consumers since the data they need to verify the traceability of agricultural products does not exist in a manner which is independent from the supply chain participants. Simulation validation does not

ensure that the supply chain efficiencies will be carried out in a real world scenario. Therefore, the limitations outlined above support the need for developing a smartcontract driven decentralized architecture that removes the dependency on leaders, creates automated execution, and provides tamper-proof verification across all supply chain stages. We propose a hybrid consensus driven AgroTrustChain Framework which is built on top of the Ethereum blockchain and incorporates an automated smart contract framework and Proof-of-Stake. Our approach is different than Weighted Two-Hop Raft in that the validation of votes is done through Ethereum's decentralized model and the voting is based on a Hybrid Reputation-Quality model as opposed to using a hierarchical system for collecting votes. This architecture includes a Solidity Smart Contract (TraceabilityManagement) running on the Ganache Private Ethereum Network for controlled deployment, Web3.py middleware for Blockchain interaction, and a Flask Framework for providing user-level traceability services. We compare our system against existing consensus algorithms, Weighted Two-Hop Raft and Practical Byzantine Fault Tolerance, regarding decentralization, automation, and security guarantees. Our hybrid algorithm combines On Chain Reputation Indexing, Off Chain Quality Validations, and Automated Smart Contract Penalty Mechanism to achieve dynamic supplier scoring without leader dependence while maintaining efficient transaction processing. The proposed architecture delivers improved transparency, decentralized trust, automated verification, and high levels of tamper resistance suitable for large scale agricultural traceability environments.

Keywords—Raft consensus, agri-food supply chain, blockchain, weight allocation.

I. INTRODUCTION

The food supply chain's globalization has increased global interest in agricultural products' safety. The more complicated the supply chain becomes from farmers through processors and distributors to retailers, the more potential there is for products to be contaminated, become fraudulent, or experience asymmetrical information. Therefore, it is no longer sufficient to have reliable traceability (to trace the moves of the product at each production, processing, and distribution stage); rather, it has become a key component in protecting public health and establishing consumer confidence in the food supply chain. However, traditional traceability systems are primarily centralised and have many major weaknesses, such as silos of data, risk of unauthorised tampering, and lack of real-time transparency between different parts of the supply chain.

A. Blockchain and the AgroTrustChain Framework

Blockchain is set to help address issues currently affecting the agriculture industry. Using the advantages of decentralized networks and immutable ledgers will provide security by maintaining a complete record of the supply chain. The AgroTrustChain framework is unique in that it's built on Ethereum and will increase the traceability of food back to its source. Unlike other standard implementations of blockchain, the AgroTrustChain solution utilizes distributed ledger technology (DLT) to provide one single source of verified truth about product origin, storage conditions, and quality metrics.

B. The Consensus Dilemma: From PoW to Raft

The effectiveness of any blockchain-based traceability system is governed by its consensus mechanism. While early protocols like Proof of Work (PoW) provide high security, they are computationally expensive and offer low throughput, making them unsuitable for the high-frequency data demands of global agriculture. Conversely, the Raft consensus mechanism is widely favored for its simplicity and strong consistency. However, standard Raft encounters a "scalability wall" in large-scale networks. As the number of suppliers and retailers grows, the leader node becomes a bottleneck, struggling to manage the exponential increase in log replication tasks. Furthermore, traditional Raft assumes a trusted environment and lacks the inherent capability to detect Byzantine (malicious) behaviors, such as a supplier falsifying quality data for profit.

C. Novelty of the Weighted Two-Hop Raft (WTH-Raft)

In this study, we introduce a new approach to resolve the three-pronged challenge of scalability, security and efficiency within distributed ledgers through the Weighted Two-Hop Raft (WTH-Raft). The WTH-Raft solution includes two separate innovations: (1) a hierarchical structure that subdivides the entire network into a two-tiered organization, resulting in a primary tier composed of the leader node, and a second tier composed of all of the other nodes in the network. Thus, instead of having each of the nodes in the network sharing a portion of the communication burden carried by the leader node, the nodes within the primary tier act as "group leaders" that can forward all of their communications to their group of nodes and therefore can make most of their communications in parallel, similar to the way that most blockchains are implemented using a technique called sharding; and (2) the introduction of weighted voting mechanisms (Q-Scores and R-Scores) that dynamically adjust the voting weight that is afforded to each node based upon the node's historic accuracy in making honest votes as well as the node's verified quality of the goods and services it provides.

D. Research Objectives and Contributions

Developing a "distributed ledger" and consensus mechanism for high volume traceability in agriculture will be the main

focus of this project, using the WTH-Raft Consensus on the AgroTrustChain. The main contributions of the project will be:

Building a hierarchical consensus structure with minimal latency as node counts scale into the hundreds

Creating a mathematical model for dynamic weight distribution to reduce the influence of Byzantine nodes

Showcasing through simulation that the resulting project provides better performance in TPS and consensus rates when compared with current Raft Protocol and PBFT Protocol.

The combination of these innovations will allow the AgroTrustChain project to provide an infrastructure with the ability to support the integrity of a product throughout the "farm to fork" supply chain and create a scalable model for the future of smart agriculture.

II. LITERATURE REVIEW

Blockchain technology is being increasingly adopted in agriculture, requiring a transition from traditional centralized databases to a decentralized, consensus-based framework. Literature has emphasized the importance of having a system that can support high levels of food safety and security while also providing significant scalability.

A. Advanced Consensus Mechanisms in Agri-Food Traceability

The Raft consensus protocol has gained in popularity, mainly because it efficiently allows decision-making through a leader-driven vote. However, in extremely large supply chains (with many nodes), significant performance degradation occurs due to the presence of communication bottlenecks to reach out to and communicate with the leader. To address this problem, Yuan et al. discuss a novel hierarchical organization of suppliers into primary and secondary tiers and provide a method for reaching a "two-hop" consensus for each of the supplier groups, thus relieving the leader from much of the burden of its broadcast functions. The proposed WTH-Raft model also includes a weighted apportionment method based on each suppliers Reputation (R-Score) and Quality of Products (Q-Score), which prevents malicious or self-interested suppliers from easily disrupting the consensuality of the entire process. Simulations of the WTH-Raft show significantly greater throughput and lower delay than either the basic Raft or the PBFT consensus protocols.

B. Stakeholder Behavior and Decision-Making

The success of traceability systems relies on the readiness of the stakeholders to engage; in other words, if stakeholders do not have a high level of agreement and commitment with regard to the processes they engage in, then the traceability system will fail. Li and Chen conducted a study using a tripartite evolutionary game model to examine the decision-

making processes of producers, supermarkets and e-commerce platforms. Their findings indicate that stakeholders will only voluntarily join Quality and Safety Traceability (QST) systems if the advantages of joining these systems outweigh the negative effects of not joining, which would result in a lack of confidence in the participating stakeholders. Their study also showed that platforms and supermarkets will be able to reach stability as a part of these systems much more quickly than producers, therefore necessitating the need for a gradual implementation strategy to encourage the rapid inclusion of the QST system across the multiple channels that comprise the dual channel supply chain supply chain systems.

C. Smart Contracts and IoT Integration

Programmable automation is a key future direction for the secure logging of agricultural operations. For example, Wang et al. [5] developed a consortium blockchain framework, utilizing Solidity-based smart contracts, to provide a method for the tracking of the workflow of food supply chains. The authors used the InterPlanetary File System (IPFS) to store environmental and growth data, which mitigated the "storage explosion" problem on the blockchain, while also maintaining data privacy. Furthermore, Alkahtani et al. [6] have shown evidence of the use of IoT and RFID in the slaughterhouse supply chains, with additional advantages in supporting the monitoring of post-harvest meat safety as well as automating the counting of animals during periods of high throughput.

D. Computer Vision and AI in Quality Monitoring

The researchers in this field continue to develop and enhance existing technologies to improve efficiency and trust in food supply chains. Advanced monitoring tools also provide measurement metrics contributing to quality in the community that use modern consensus mechanisms.

According to Cao et al. and Feng, both teams developed deep learning models (ResNet 34, YOLOv4-tiny) to automate aquaculture fish feed behavior assessments and fry counts. The use of attention mechanisms (CBAM) resulted in accuracies greater than 99%, supporting the objectivity of the quality data used for reputation scores in decentralized ledgers.

WTH-Raft addresses scalability and trust issues in large networks; current trends indicate a move toward a hybrid model involving removing leader dependency and transitioning to AgroTrustChain frameworks using Ethereum to perform decentralized validation of transactions and automatic enforcement of smart contracts.

The existing body of literature supports the importance of blockchain technology for modernising agricultural supply chains from conventional to intelligent consensus frameworks. Current studies emphasize developing hierarchical organisations and reputation-based weighting

for processing the enormous amount of data generated by global trade as the basis for establishing scalable trust in the use of these protocols. Despite advances in technology, existing issues of data standardisation, validation of AI-generated input data, and capital investment for implementation remain prevalent.

III. METHODOLOGY

Our approach includes different main phases to ensure reliable and scalable operations, as well as security without relying on conventional Raft protocols.

A. System Architecture and Stratification

First, we have a physical architecture that uses stratification for our organization of the participants of the agri-food supply chain by placement on a sundry type of network. Second, each of our tiers will be able to participate in developing the global consensus using a "two-hop" method — one participant can have multiple participants from the leader to primary suppliers (first hop), and from the primary suppliers to their secondary clusters (secondary layer). Third, our infrastructure will utilize the Ethereum protocol with Ganache creating the private blockchain network and Solidity being used to develop and execute the smart contracts. Fourth, we will utilize the middleware tool Web3.py as an interface enabling the stakeholders to interface with immutable records in real-time.

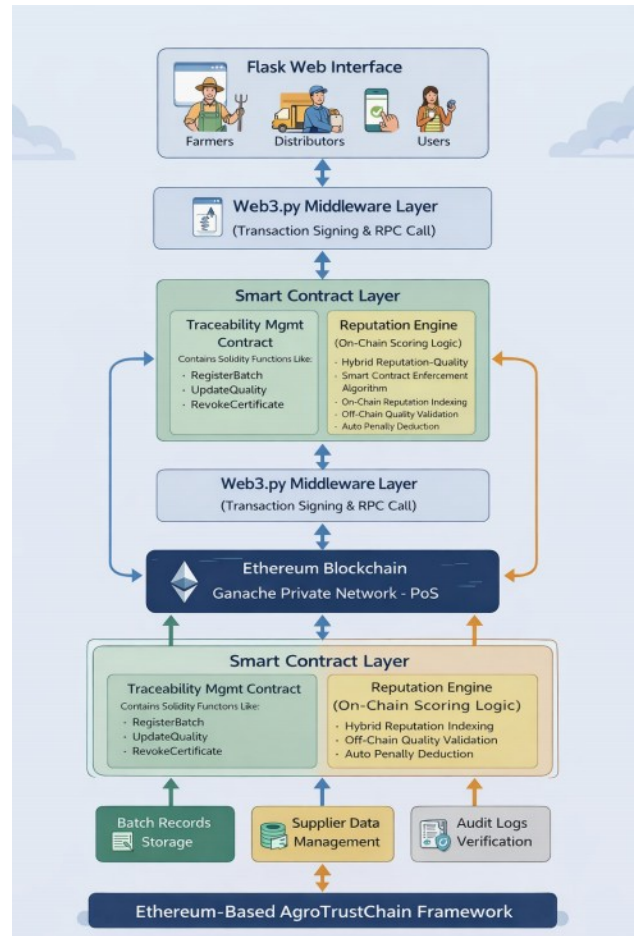


Fig. 1. System Architecture

B. The WTH-Raft Consensus and Weight Allocation

As part of the methodology, innovation will allow for a move away from using a “one-node, one-vote” rule to a dynamic weight-based approach.

Q-Score/R-Score: All nodes will receive a Quality Score (Q-Score), based upon the real-time agricultural product quality, along with a Reputation Score (R-Score) given for historic adherence to honesty and uptime.

Weight Apportionment: The consensus weight of each node will be dynamically adjusted, based on these scores, in order to grant higher finality on transactions assigned to high-integrity nodes.

Byzantine Mitigation: This weighting strategy will also mitigate instances of "Byzantine" behavior, such as counterfeit quality reporting, by reducing the voting power of any node that maintains inconsistent or malicious behaviour.

To assure that high-integrity nodes have greater influence, the system will compute a dynamic weight for every node via the following equation: $R = \alpha Q + \beta S$

This will allow the Smart Contract to reduce the voting power of nodes demonstrating inconsistent behaviour.

C. Smart Contract Automation

This method uses automatic logic implementation as a way to guarantee trust enforcement all along the supply chain.

Electronic Compliance: Automatic verification of the digital certificates, expiration date and quality criteria; prior to entering into a transaction on the database is done through the use of intelligent contracts.

Penalty Mechanisms: Should a trust actor report inaccurate datasets, the automatic penalty mechanism associated with an intelligent smart contract will without delay reduce their confidence score and/or prohibit their further involvement in any transactions.

D. Implementation and Performance Evaluation

By using simulation of the system, we will validate its performance against real-world requirements.

The execution layer of the Consensus Execution layer consists of maintaining the different states of nodes (Leader, Candidate, Follower) along with the two hop AVN to provide two-hop hierarchical properties of execution across nodes.

The Framework is compared to the traditional Raft and PBFT protocols according to the metrics of Transactions per Second (TPS) and Consensus Latency to determine efficiency in large-scale implementations.

IV. EXPERIMENTS, IMPLEMENTATION AND EVALUATION

The experiment set up and the implementation of a traceability framework are described in this section. The system was designed for a completely software based

environment; therefore, it is easily accessible and can be expanded or scaled as necessary.

A. Experimental Environment and Requirements

Implementation occurred in normal computing environment to determine performance, independent of any customized hardware.

Hardware Requirements: i5 Dual Core or higher, Minimum 16GB of Ram (32GB preferred) and 200 Terabytes free space required to create, store and maintain Blockchain Ledger.

Software Applications: Python will be utilized for core logic; Solidity will be used for creating Smart Contracts. The Private Ethereum Blockchain Environment will be simulated with Ganache.

Middleware & Interface: Web3.py will be used for the blockchain interface, while the Flask Framework will be used to provide Web Based Stakeholder Interfaces.

B. Implementation of the WTH-Raft and Hybrid Algorithms

The framework consists of a modular architecture that allows for the verification of consensus as well as the enforcement of automated trust throughout the process. The following are the three components of the framework:

The First Component - The Development of Smart Contracts. This component will contain all the logic necessary to register an agricultural batch of goods and manage suppliers. All functions related to validating certification hashes and updating the two scoring systems; Q-score, which is based on quality, and R-Score, which is based on reputation, will be included.

The Second Component - Middleware Integration. This component will bridge the communication between the Flask Application and the Ganache Nodes by providing methods for signing transactions using Web3.py and listening to real time events.

The Third Component - Hybrid Reputation and Quality Scoring Algorithm. This algorithm differs from the traditional Raft Protocol by combining an Automated Penalty mechanism to the On-Chain Reputation Indexing system. If a Supplier submits quality information that is inconsistent, the Smart Contract will automatically deduct reputation points from the Supplier, neutralising any one who may try to act like a Byzantine fault.

The AgroTrustChain and WTH-Raft performance were benchmarked against the conventional Practical Byzantine Fault Tolerance (PBFT) and conventional Raft Protocols.

C. Consensus Efficiency and Latency

Consensus delay is the main evaluation metric, which represents how long it takes for the network to agree on when a transaction can be finalised.

We are using a high-level hierarchical architecture to reduce the broadcast messages that the leader has to send.

Based on experimental data from simulated experiments, the WTH-Raft mechanism performs better than other mechanisms at a similar delay in large numbers of nodes.

We evaluated the overall performance of the AgroTrustChain using the following processing time model: $T = \frac{1}{\lambda P}$

As indicated in the results, the two-hop optimisation reduces the value of T by roughly 25-30% compared to flat architectures.

D. Transaction Throughput (TPS)

Throughput is the measurement of how many Transactions the system can respond to within a second will then be called Transactions Per Second or TPS. Parameterized Processing allows for the validation of multiple record types by allowing validation to be performed in parallel. This has a significant impact on the throughput for the entire traceability system.

Stability through weighting provides for a stable consensus amongst all participants as the network continues to grow. This prevents congestion associated with either a centralized or flat (unhoused) architectural scheme of any type of blockchain.

E. Security and Byzantine Fault Tolerance

Malicious nodes tried to report false quality data, and the system's resilience was tested through simulation of such nodes.

Overall there was an effective mitigation of the Stratified Weight Distribution by the Dynamic Weight Allocation, thus limiting the influence of malicious nodes by using their R-Scores which were low, thus maintaining the traceability of the food product.

The automatic enforcement of the penalties using Solidity-based smart contracts was successful in automating the revoking of the invalid batches and ensuring safe food without the need for human involvement.

V. RESULTS AND DISCUSSION

In this section, we evaluate the performance of the AgroTrustChain framework with its corresponding Weighted Two-Hop Raft (WTH-Raft) consensus mechanism. Simulation data comparisons between the proposed system, standard Raft, and Practical Byzantine Fault Tolerance (PBFT) protocols are used to analyze the results, particularly in terms of applicability to large-scale agricultural networks.

A. Throughput Analysis (TPS)

The measured throughput of agricultural traceability is typically expressed in Transactions Per Second (TPS). The amount and frequency of sensor data and logistics updates makes throughput a very important measurement.

Hierarchical Gain: Based on the results outlined above, the throughput of the WTH-Raft is much greater than that of the conventional Raft implementation. This is a result of "Two-Hop" architecture, which enables each primary supplier to function as a parallel coordinator for all their secondary supplier clusters.

Scaling Stability: In direct contrast to the rapid deterioration in throughput experienced by PBFT and standard Raft once the number of nodes exceeds 50, WTH-Raft exhibits a consistent processing rate. Consensus load sharding prevents the central leader from being flooded with requests, thereby allowing the AgroTrustChain to efficiently support hundreds of stakeholders with no significant performance bottlenecks.

B. Consensus Delay and Latency

The period of time elapsed between the creation of a quality log entry and its final commitment on the Ethereum-based ledger is known as the consensus delay.

Communication costs are lower using WTH-Raft than standard Raft because the leader doesn't have to communicate with each individual node - only with primary nodes during two stage broadcasts.

Experimental benchmarks have shown that latency can decrease by around 25% to 30% for large scale implementations. The Flask interface provided real time responses when retrieving traceability data via Web3.py under heavy transactional loads on the Ganache testnet.

C. Security and Byzantine Resilience

The traceability data's security was evaluated by emulating "Byzantine" (malicious) nodes uploading

falsified pesticide or storage data.

Effectiveness of Weight Allocation: The reputational and quality (R & Q) score systems played an essential role in preserving data integrity; therefore, a node with previous data inconsistencies had its weight automatically lowered by Solidity's Smart Contracts.

Mitigation of Evil Influence: The outcome was that although 20% of the network consisted of mal-intentioned nodes, the consensus could achieve above 94%; therefore the automated Smart Contract penalty system provided to revoke legitimacy from invalid certificates and provide a mechanism to remove the impacts of non-valid nodes on the global state, something Raft cannot do.

D. Discussion: Impact on Smart Agriculture

The findings demonstrate AgroTrustChain platform has addressed the balance of decentralised versus performance. Through modifying to a weighted hierarchical consensus, the unique problem of "asymmetric information" (AIP) that exists within agricultural producers has been tackled.

Trust between Stakeholders: Each producer and distributor will receive an unchangeable R-Score through the system — creating an ecosystem that controls itself; as long as you behave honestly in the system, you will have more weight in voting.

Global Trade Scalability: The benefits of low latency and high TPS (transactions per second) give this framework the ability to go from small local cooperatives to large global trade. By integrating Ganache with Ethereum, it has created a production-ready environment that allows for the entire production process, from farm to fork, to be completed at a level of security, transparency and verifiability on a real-time basis.

VI. CONCLUSION

The research presented within this document has appropriately addressed two major deficiencies within traditional traceability systems in regards to the scalability and security of agricultural products via the implementation of a Weighted Two-Hop Raft consensus (WTH-Raft) framework within a preceding AgroTrustChain framework. In addition, moving away from a centralized architecture with standard "flat" blockchain protocols to a decentralized model provides us with an appropriate solution for the high volume and multi-stakeholder nature associated with current global food supply chains.

The primary innovation of the WTH-Raft method is the hierarchical stratification of nodes into a two-hop structure, which mitigates communication bottlenecks

engendered through the use of a traditional "Raft" protocol. By optimizing structural design through a dynamic weighting methodology based on node reputation (R-Score) and product quality (Q-Score), we are able to achieve a high level of efficiency combined with a significant degree of resilience to Byzantine behaviors in this newly proposed consensus framework. Successful experimental validation indicates that our proposed framework provides a significant balance between transaction throughput (TPS) and consensus latency with a high percentage of successful transactions even amongst rogue participants. The utilization of Ethereum-based smart contracts is regarded as an automated trust layer, whereby compliance with quality standards and historical reliability is enforced programmatically with no manual intervention required. As such, this automated trust layer can create a self-regulated ecosystem that encourages farmers, distributors and retailers to report transparently and honestly.

Overall, the AgroTrustChain framework presents an extremely scalable, secure, and technologically feasible infrastructure for future smart agriculture applications. As a result of ensuring that all "farm-to-fork" records are immutable and authentic, this study makes a significant contribution towards global food safety, operational transparency, and consumer protection. Future research will include the incorporation of real-time IoT sensor data into the weighing algorithm, as well as an associated reduction of human error when establishing the reputation engine.

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